

A SHORT COMPANION TO DOUBLE-SLIT GEOMETRY

Euclid's Five, Revisited

The founding brief asked for Euclidean geometry with a rewritten fifth postulate. This document pays the original debt: the five postulates, one at a time, each with its exact fate in the geometry of the double slit — and each fate verified numerically (`check_euclids_five.js` , 20 assertions).

0 THE PLANE WE ARE STANDING IN

Euclid's postulates are statements about a plane, so we must say which plane. It is the **image geometry** of the double slit: the ray space $P^1 \times P^1$, whose points are rays, whose "lines" are the chains, and whose coordinates are the double numbers $\mathbb{R} \oplus \mathbb{R}$. The incidence postulates (1, 2, 5) are read at Level 1 of the hierarchy — pure incidence of rays, pencils, and chains. The metric postulates (3, 4) are read in the tangent plane of that geometry, where the double numbers supply their own norm, $N(\delta) = \Delta s \cdot \Delta t$ — the split norm whose zero-set is the seam. Every claim below is a theorem of that structure, and every claim below has a passing assertion in the check script.

E1 THE FIRST POSTULATE

To draw a straight line from any point to any point.

FATE: THE COUNT SHIFTS BY ONE

Through two generic rays pass **infinitely many** chains; through two parallel rays pass **none**; it takes **three** pairwise non-parallel rays to determine exactly one (Axiom M1). The "lines" of this geometry behave the way circles behave in inversive geometry — which is what they are.

Verified: two distinct chains constructed through the same pair of rays; any attempted chain through a 1-parallel pair

degenerates ($ad - bc = 0$); three-point determination confirmed.

There is a consolation for the parallel pair: though no chain joins them, they lie together on a pencil — the degenerate lines of the geometry, and the arms of the crosses of Theorem 3. Euclid's postulate does not fail so much as fork: joining two points is either a three-point affair or a seam affair.

E2 THE SECOND POSTULATE

To produce a finite straight line continuously in a straight line.

FATE: IT EXTENDS — AND RETURNS

Extension is unobstructed, but it is not endless: a chain is a **closed loop** in $P^1 \times P^1$. Followed past its pole, the chain passes through the single point at infinity of one factor and comes back along the other branch; followed off both ends of the chart, it returns to the same value. Every line of this geometry is a circle that happens to pass through the chart's corner.

Verified: through the pole, $1/t \rightarrow 0$ from both sides (one projective point ∞ , crossed once); as $s \rightarrow \pm\infty$ the chain closes onto the same $t = a/c$.

E3 THE THIRD POSTULATE

To describe a circle with any centre and radius.

FATE: GRANTED — TWICE OVER, WITH ONE DEGENERATE SURPRISE

Circles about any center exist for every radius under the split norm: $(s-c_1)(t-c_2) = k$, and each is a chain. The radii come in **two causal kinds** ($k > 0$ and $k < 0$ — Euclid's circles doubled), and the circle of radius **zero** is not a point: N vanishes on the zero-divisors, so the degenerate circle is a **cross** — the two pencil lines through the center. The seam is what a point looks like when a circle shrinks onto it.

Verified: split circles of both signs fit Möbius graphs to 10^{-15} ; the $k = 0$ locus is exactly the two pencil lines.

split circles $(s-c_1)(t-c_2) = k$ about one center

$k = 0$: the circle degenerates to the CROSS (blue/red)

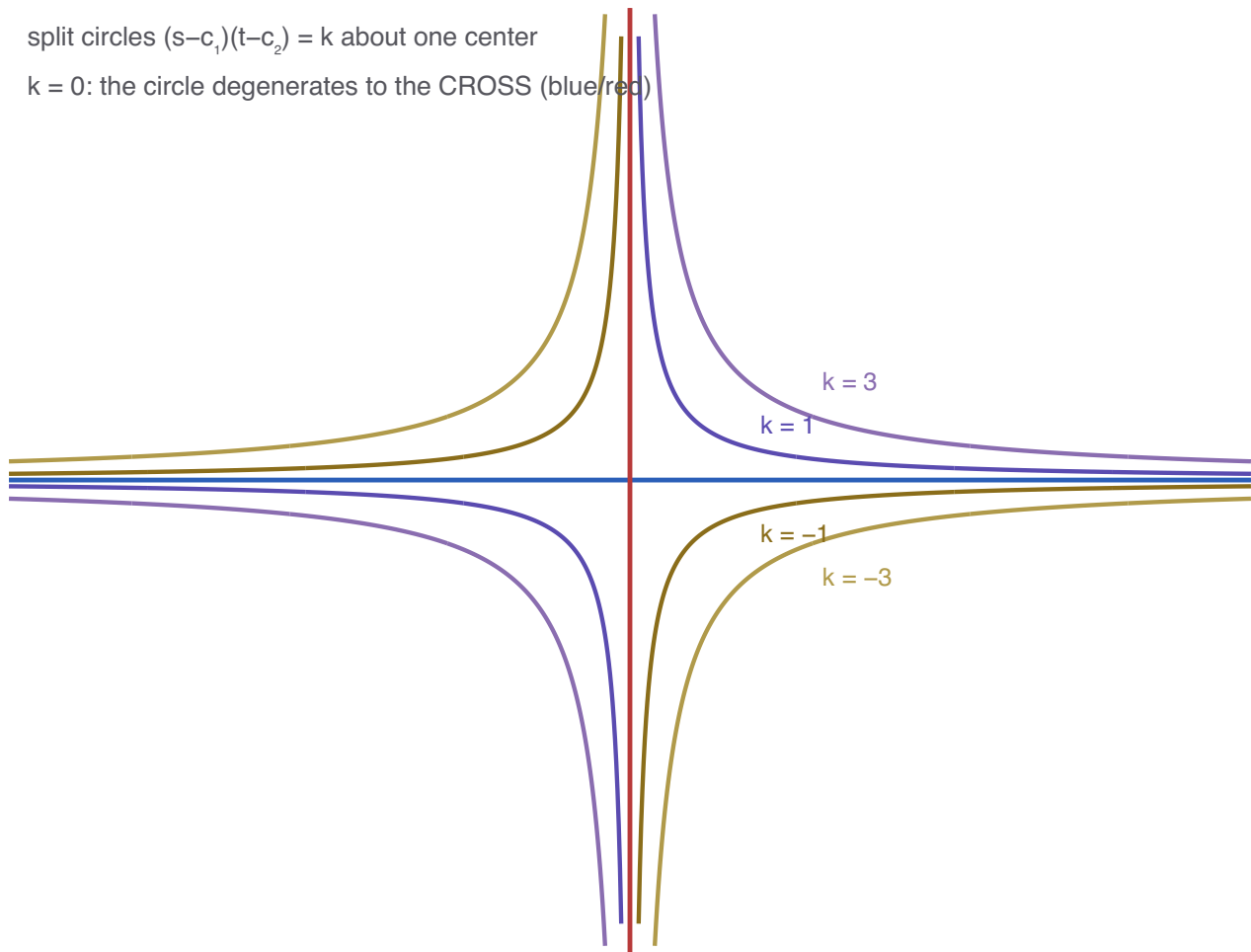


Figure A. Concentric split circles about one center, computed: violet family $k > 0$, gold family $k < 0$, and at $k = 0$ the circle degenerates onto the cross of pencil directions (blue and red). Each curve is asserted to be a chain before it is drawn.

E4 THE FOURTH POSTULATE

That all right angles are equal to one another.

FATE: THE SLEEPER — THE CONGRUENCE HOLDS, THE ROTATION BENEATH IT BREAKS

Right angles exist: the perpendicular of slope m is slope $-m$, and the null directions are perpendicular *to themselves* — the seam is its own normal. All right angles are congruent, exactly as Euclid demanded: an explicit boost carries any onto any. But the **quarter-turn does not exist**. Every isometry of the split norm preserves the sign of slope, so no motion carries a line onto its own perpendicular; the two arms of every right angle lie in different

causal sectors and are never exchanged. And the directions that bisect every right angle are the null directions: **the seam bisects every right angle** in the geometry.

Verified: polar form vanishes on $(m, -m)$ pairs; null self-perpendicularity; both $O(1,1)$ families preserve N and slope sign; explicit congruence boosts for same- and opposite-sign right angles; every slope path between the arms crosses a null direction.

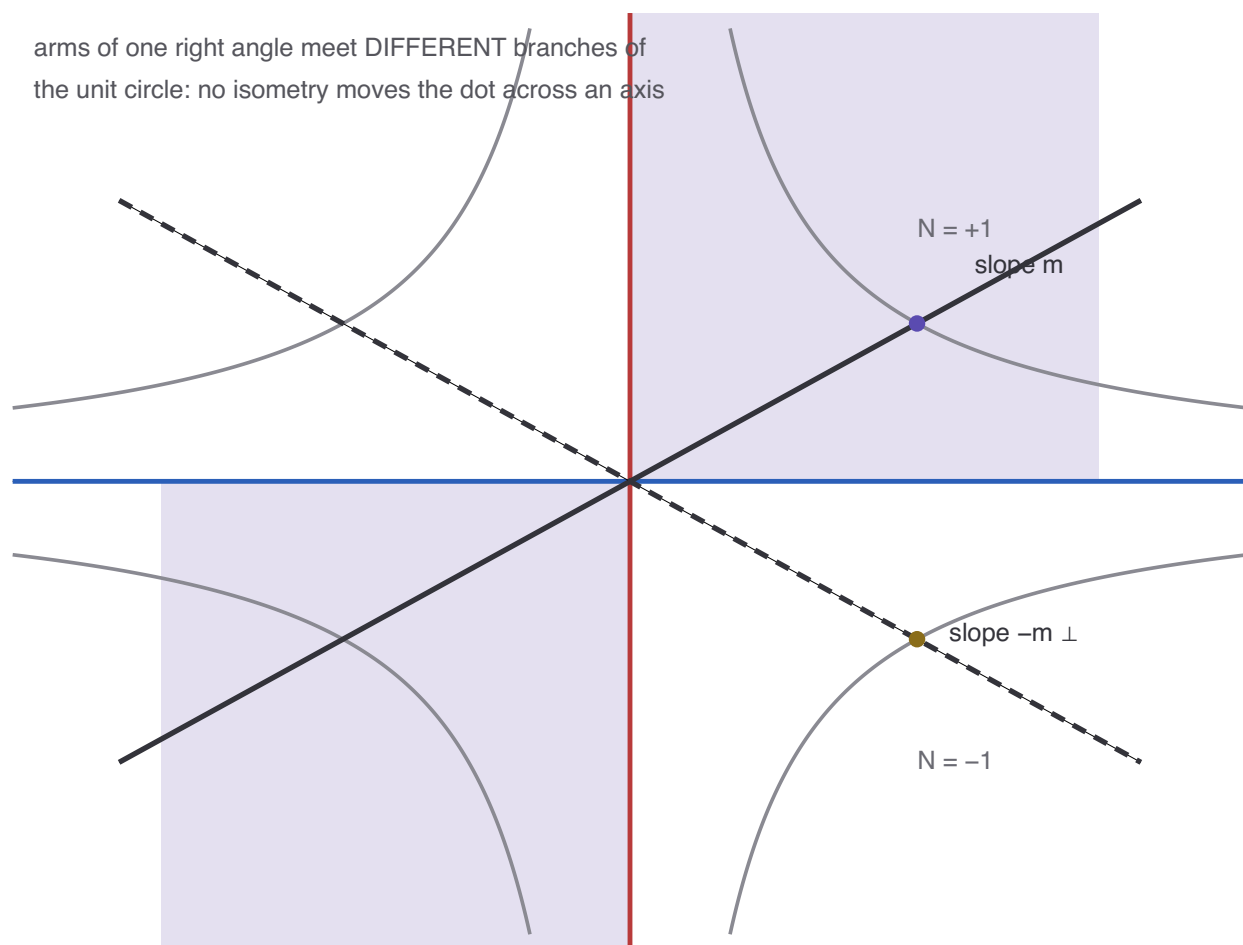


Figure B. The unit circle of the split norm is four hyperbola branches ($N = \pm 1$). The arms of one right angle — slopes m and $-m$ — meet branches of *opposite sign* (violet and gold dots), and no isometry moves a dot across an axis. What breaks in the fourth postulate is not the equality Euclid asserted but the rotation he assumed beneath it.

E5 THE FIFTH POSTULATE

That, if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side... (Playfair's form: through a point not on a line there is exactly one line that does not

meet it.)

FATE: THE QUESTION TRANSFORMS — AND THE POSTULATE SURVIVES AS TANGENCY

Parallelism itself **splits in two**: through any ray there is exactly one 1-parallel class and one 2-parallel class (Axiom M0) — Euclid V holds twice over for pencils. For chains, Playfair fails as stated: through a ray off a chain K pass *several* chains disjoint from K in all of $P^1 \times P^1$ — non-meeting is not a privileged relation here, any more than it is for circles. What carries Euclid V's structural role — one privileged line through an outside point — is the Touching Axiom M2: through a ray q off K there is **exactly one** chain meeting K precisely at a prescribed ray, and it matches K to first order there. In double-slit geometry, **parallel survives as tangent**.

Verified: three distinct chains through one ray, each disjoint from K over the whole closed surface (constructed as elliptic composites — the intersection pair driven complex); the touching chain is unique (the second intersection point is a Möbius function of the family parameter, so it takes the value of the first exactly once), and its slope matches K's at the contact ray to 10^{-6} .

The classical alternatives to Euclid V change the *answer*: no parallels (elliptic), many parallels (hyperbolic). The double slit changes the *question*: disjointness stops being the interesting relation, and tangency inherits the uniqueness. This is the fifth postulate's fate in every circle geometry, and the Minkowski plane is no exception.

Σ THE FIVE AT A GLANCE

Euclid	Double-slit fate	Carried by
1. A line through any two points	Two rays: many chains, or none (parallel pair); three rays: exactly one	M0, M1
2. A line extends indefinitely	It extends and returns : every chain is a closed loop	D3, Thm 2
3. A circle with any center and radius	Granted, with radii of two causal kinds; radius zero is a cross	$N = \Delta s \cdot \Delta t$
4. All right angles are congruent	Congruent, yes — but the quarter-turn does not exist , and the seam bisects every right angle	$O(1,1)$
5. The parallel postulate	Parallelism doubles ($\parallel_1, \parallel_2$); Playfair fails for chains; uniqueness	M0, M2

The punchline the project earned rather than assumed: the brief set out to rewrite the fifth postulate. The finished geometry mostly *keeps* the fifth — doubled at the pencils, transposed into tangency at the chains — and it is the **fourth** that breaks, at the rotation Euclid never thought to postulate because in his plane it was free.

R REFERENCES

1. W. Benz, *Vorlesungen über Geometrie der Algebren*, Springer, 1973. (Minkowski planes; chains as circles.)
2. I. M. Yaglom, *A Simple Non-Euclidean Geometry and Its Physical Basis*, Springer, 1979. (The split plane's trigonometry: boosts for rotations, the null cone as angular horizon.)
3. F. Klein, "Vergleichende Betrachtungen über neuere geometrische Forschungen" (the Erlangen program), 1872.
4. Project documents: [Double-Slit Geometry: Definitions, Axioms, and Theorems](#) (the axioms M0–M4 and Theorems 1–8 cited throughout); [The Seam](#) (the double numbers and the null structure); [Addenda](#) (A9: the reordering orbit and the harmonic collapse on the diagonal).

Companion to **Double-Slit Geometry: Definitions, Axioms, and Theorems**. Every fate stated above is asserted by **check_euclids_five.js** (node, no dependencies): 20 checks covering both figures and all five postulates. Rig conventions as in the main document; the split norm and its isometries are taken in the tangent plane of ray space.